

Task-Oriented Deep Joint Source–Channel Coding

What a camera should send when the receiver only needs the answer



THESIS

Train the sender and the receiver together so the classification decision survives a short, noisy link.

Second Review, 29 September – 3 October 2026.
Everything shown here was measured on the validation split. No test data has been read.

What changed between the two reviews

In August we had a working classical chain and no trained learned model. Both exist now.

Aug	Classical baseline	3,213 measured BLER identities, 153 BLER curves, 288,000-row validation sweep, per-SNR adaptive selection frozen
Aug	Learned training	Checkpoint/resume loop, multi-seed headline models, three frozen seed cells at 1/6
Sep	Attribution control	ER-9 task-aware digital features over the same LDPC chain — the control for H4
Sep	Robustness	SNR-randomised training variant, compared against fixed-SNR training
Sep	Crossover decision	G-10 closed on frozen validation curves: crossover observed between -5 and -4 dB
Sep	Peak-power constraint	One PAPR-constrained training run, capped at 3 dB in the symbol domain
Oct	Validation rehearsal	252 measurement units across 12 arms and 21 SNRs, 1,000 images each
Nov	Test campaign	Still sealed. Opens at G-12 once the freeze manifest is signed

Everything above ran on the validation split. The test split stays sealed until gate G-12.

Objectives, stated as completion criteria

The rubric asks whether objectives were met, so they are written as things we did, not results we hoped for.

WE SET OUT TO

- 1 Build** A learned encoder and decoder trained end to end for image classification over a simulated AWGN link.
- 2 Bandwidth-match** Every system gets the same channel-use budget at each SNR and ratio.
- 3 Bit-account** Filler, CRC, code-block segmentation and rate matching are charged before any source byte.
- 4 Evaluate** At the operating point chosen by a classical-only rule, so the selection never saw the learned curve.
- 5 Report** Paired per-image outcomes, including the outage decision, on every row.

NOT THE OBJECTIVE

"Show that the learned system beats the classical one."

That is a result, not an objective. A crossover and learned dominance are both complete outcomes under SPEC §2, and the rubric scores the second review on objectives we set — not on which way the curve went.

G-10 found a crossover, so DEC-16's fallback clause never fired and the objectives were not rewritten.

What was measured for this review

252

measurement units, each
one row of the published
aggregate

12

system arms, including five
controls

21

SNR points, -8 dB to +18
dB on the measured grid

1,000

validation images behind
every point

0

test images read, at any
point, ever

BANDWIDTH

1/6 ratio — 12,800 channel uses per image
1/24 ratio — 3,200 channel uses per image

These are matched channel-use budgets within a ratio. They do not match transmitted information, and the learned and digital arms carry different representations.

SCORERS

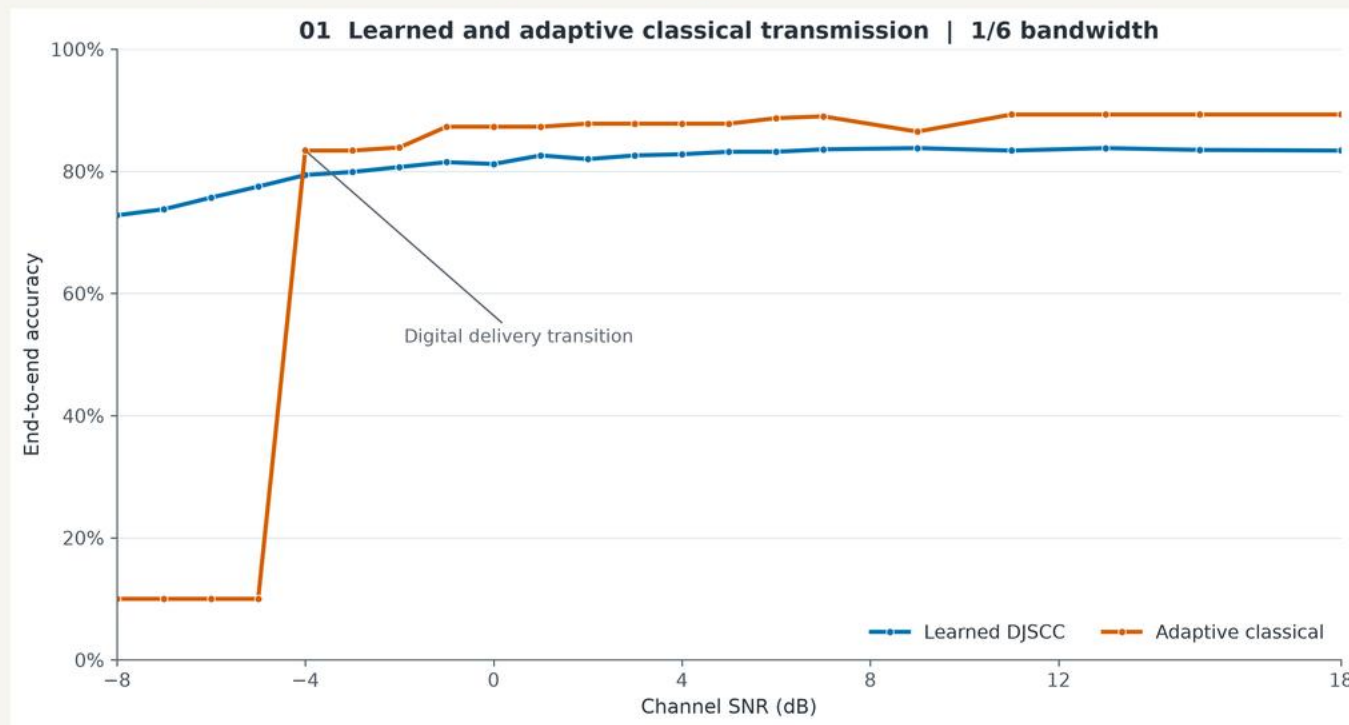
Learned arms are scored by their own task head. The classical arms are scored by the artifact-finetuned classifier; the same transmitted outputs also carry a clean-classifier stream in the published scorer CSV.

So an end-to-end learned-versus-classical gap combines communication, representation and scorer. We treat it as a whole-system comparison and say so wherever it appears.

One frozen seed cell. These are observed point differences, not confidence intervals.

Learned DJSCC against the adaptive classical chain

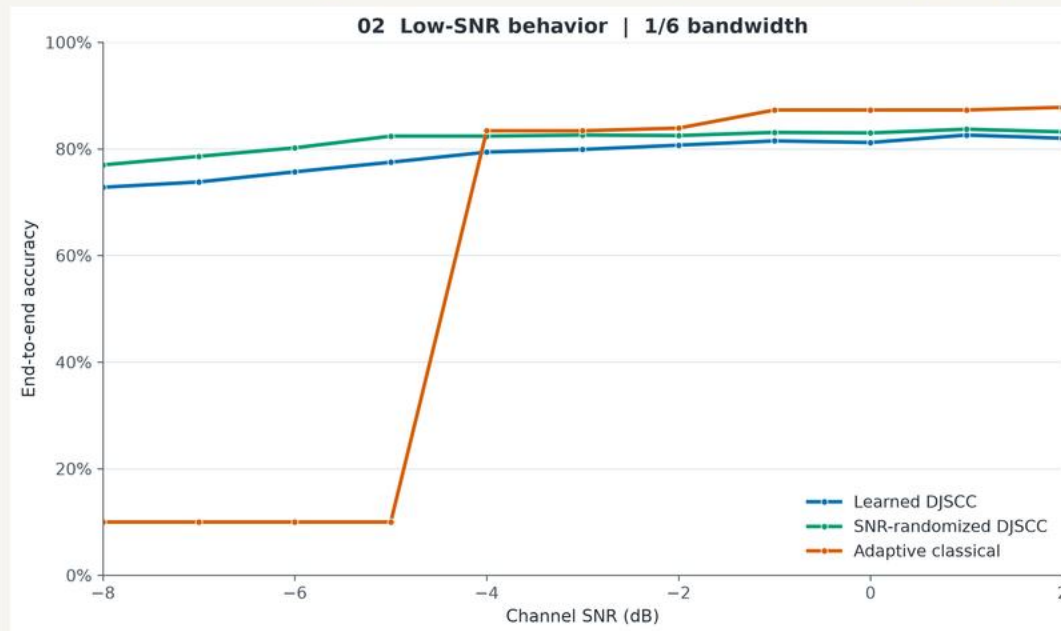
The learned system answers at every SNR. The digital chain is silent until -4 dB, then overtakes it.



At -8 dB: learned 72.8%, digital 10.0% (no delivery). At -4 dB: digital 83.4%, learned 79.4%. At $+18$ dB: digital 89.3%, learned 83.4%.

Close-up: where the digital chain gives up

Between -5 and -4 dB the digital chain starts delivering and its accuracy jumps 73 points.



-8 dB

Learned 72.8%. Digital coverage is 0%, so its score is the 10.0% outage fallback.

-4 dB

Digital coverage reaches 100% and accuracy 83.4%, above the learned 79.4%.

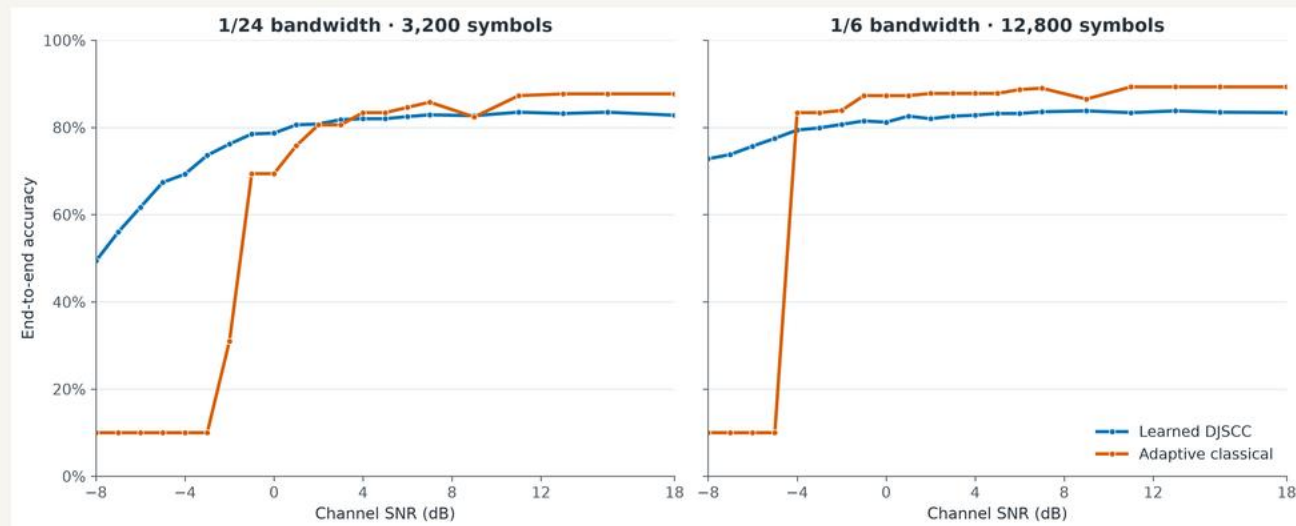
Above

Coverage dips to 97.7% at -2 dB and 96.2% at $+9$ dB, so the digital curve is not monotone.

No SNR between -5 and -4 dB was measured, so we quote a bracket and not a threshold.

Cutting the budget fourfold

At 1/24 the digital arm loses far more at low SNR than the learned arm does.



-8 dB, 1/6 → 1/24

Learned drops 72.8% → 49.4%, a 23.4 point loss.

0 dB, 1/24

Learned 78.7% against digital 69.4%.

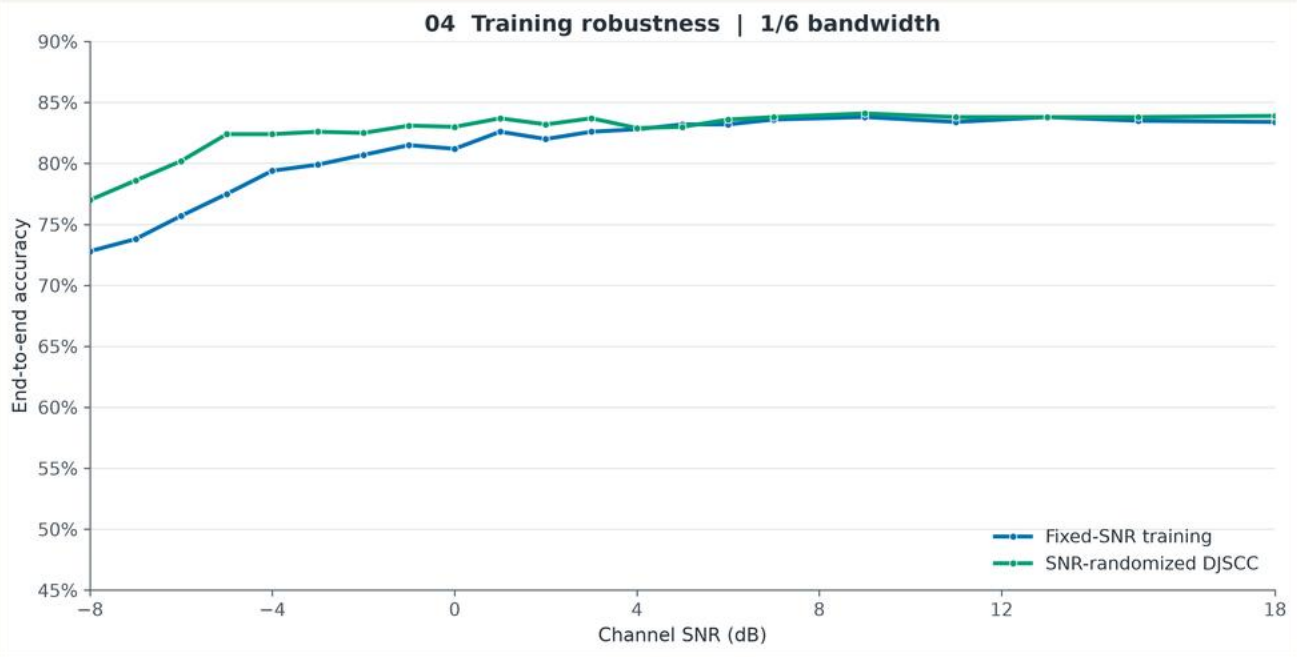
First overtaking

Digital passes learned at the measured +4 dB point, then learned retakes the lead at +9 dB where digital coverage dips.

The +9 dB recross means we should not claim one clean crossover at this ratio.

Training across SNRs instead of at one

Drawing the training SNR uniformly lifts the whole weak-channel end of the curve.



SYSTEM	-8 dB	-4 dB	0 dB	+9 dB	+18 dB
SNR-randomised	77.0	82.4	83.0	84.1	83.9

Fixed-SNR

Two separately trained checkpoints in one seed cell. This is not an estimate of seed-to-seed training variance.

77.8

79.4

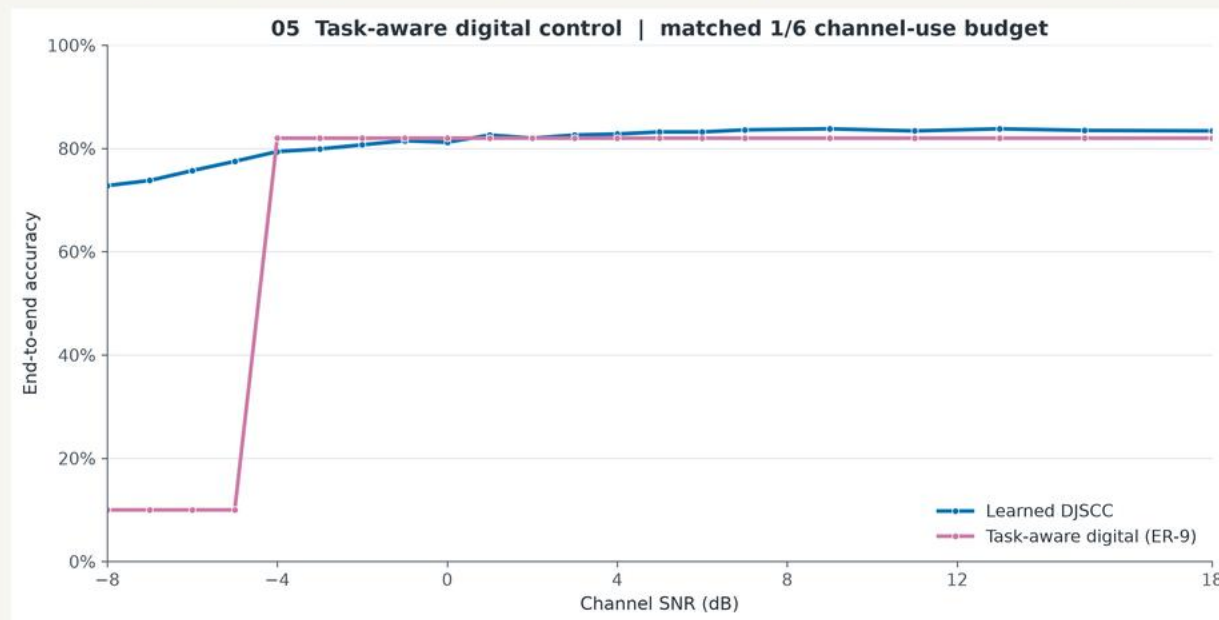
81.2

83.8

83.4

The control that keeps the credit honest

If we had only compared against JPEG, the low-SNR story would have been too easy to believe.



What ER-9 does

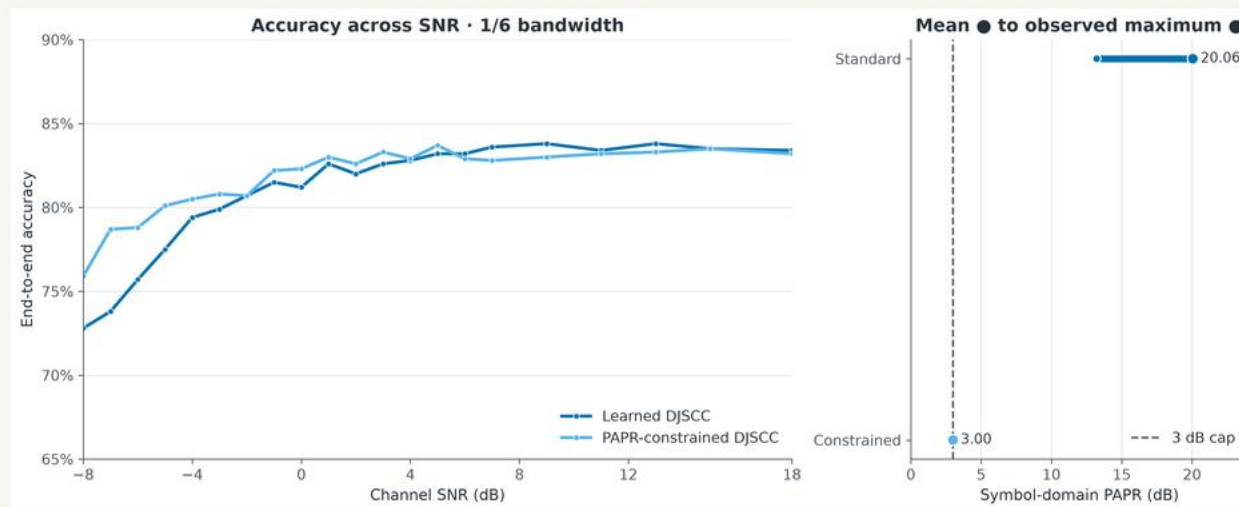
Sends learned task features over the same LDPC and QPSK chain at the same 12,800 channel uses. It is the control that separates task-aware representation from joint coding.

What it settles

It holds 82.0% at every SNR from -4 dB up. Digital transmission of task features is strong once it delivers, so the low-SNR gap is about outage behaviour as well as representation. It does not isolate the LDPC code.

Holding the crest factor under a hard cap

A PA-friendly transmitter is a real design constraint, so we trained under one.



3.000002 dB

max PAPR, constrained run — inside the frozen cap

20.059284 dB

max PAPR, unconstrained learned symbols

+3.1 pp

accuracy gain at -8 dB, constrained versus unconstrained

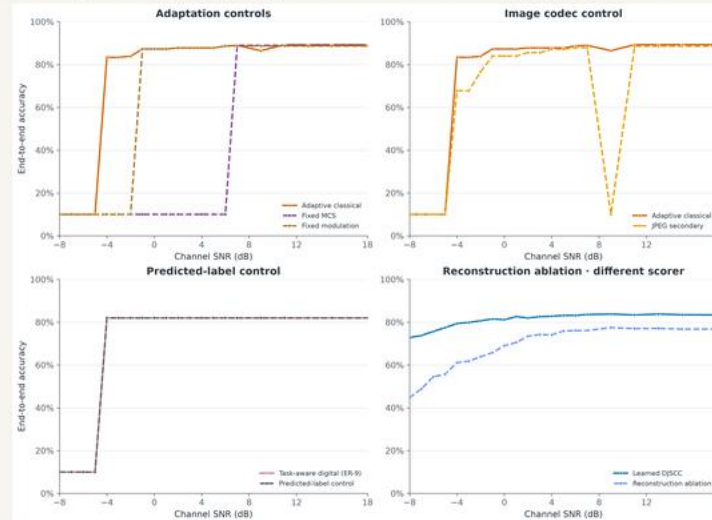
-0.2 pp

at +18 dB — the cap costs nothing up here

This is PAPR measured on channel symbols. It says nothing about amplifier or over-the-air compliance, and the constrained run is a separately trained checkpoint, so the accuracy differences are not a clean causal penalty.

Four controls, including one that went against us

Each panel takes one design choice away from the classical chain and shows what it was worth.



Fixed MCS

Outage floor until +6 dB, then 89.0%. Per-SNR selection buys real reach.

Fixed modulation

Delivers later than adaptive selection, plateaus near 88.7%.

JPEG secondary

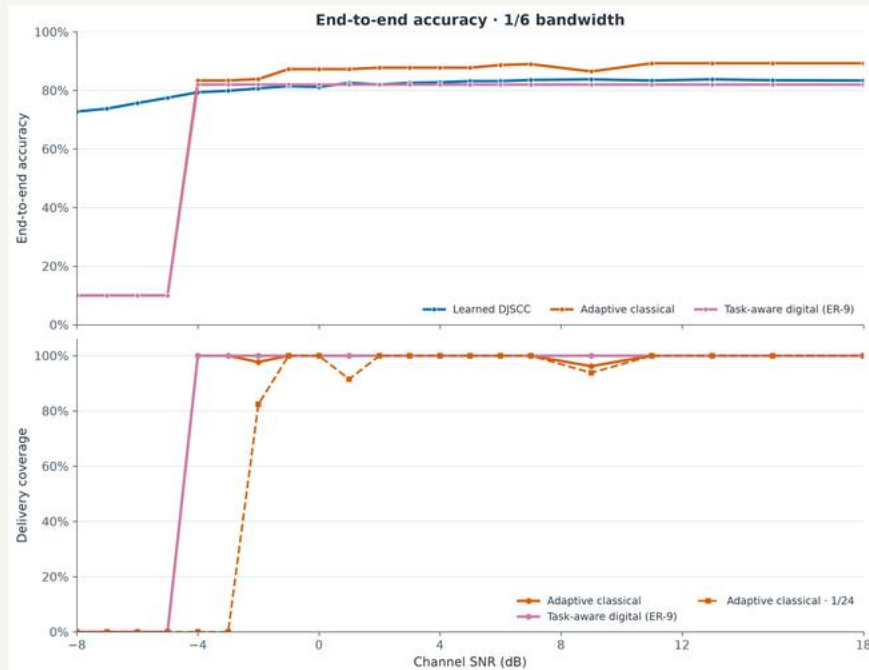
88.6% at +18 dB, and a genuine 0% coverage / 10.0% point at +9 dB. Kept in.

Predicted label

82.0% once it delivers. A transmitted label, not the oracle true label.

Why the digital curve breaks where it does

Accuracy steps track delivery steps, one for one.



READING IT

The 10.0% floor is a policy, not a delivery

When the digital chain cannot deliver, every image falls back to the single class selected on validation. On an exactly stratified split that is 100 of 1,000 images, so 10.0%.

Outage is part of the result

Coverage and accuracy are reported side by side on every row, so a reader can separate "the link failed" from "the classifier was wrong."

Learned output availability is contractual

The learned contract returns a label at every SNR, which is why its curve has no floor to sit on.

No points were inserted between measured SNRs, and no curve was smoothed to hide a step.

The numbers behind the figures

End-to-end accuracy on the frozen validation split, counted per image including outage decisions.

1/6 BANDWIDTH · 12,800 CHANNEL USES

SYSTEM	-8	-5	-4	0	+9	+18
Learned DJSCC	72.8	77.5	79.4	81.2	83.8	83.4
Adaptive classical	10.0	10.0	83.4	87.3	86.5	89.3
SNR-randomised	77.0	82.4	82.4	83.0	84.1	83.9
ER-9 digital	10.0	10.0	82.0	82.0	82.0	82.0
PAPR-constrained	75.9	80.1	80.5	82.3	83.0	83.2

1/24 BANDWIDTH · 3,200 CHANNEL USES

SYSTEM	-8	-4	-2	0	+4	+18
Learned DJSCC	49.4	69.3	76.2	78.7	82.0	82.8
Adaptive classical	10.0	10.0	31.0	69.4	83.4	87.7

G-10, the crossover decision

The measured sign of the learned-minus-classical gap flips between -5 dB and -4 dB at 1/6. That is an expected-direction crossover and it closes gate G-10.

WHAT THE TABLE IS NOT

These are 21 SNR points on one grid, not 21 independent replications. Every cell comes from the same 1,000-image validation set at one frozen seed cell, so the smoothness of a curve carries no statistical information. A one- or two-point difference between rows should be described as an observed difference and nothing more. The bulk per-image rows that would support paired uncertainty intervals are not in this checkout, so we have not computed any.

What this evidence cannot tell you

One seed cell, validation only

Every number here is descriptive. No population interval, no significance claim, no replication count.

Different scorers on the two sides

The learned arms use their own task head, the classical arms the artifact-finetuned classifier. The gap is a system-level quantity, not a controlled test of channel coding.

Not equal optimisation effort

The classical selection rules and artifact scorer were frozen from validation work; the learned controls are separate training runs. Specific and fair, but not matched compute.

No waveform or amplifier data

PAPR is measured on channel symbols. Nothing here establishes RF behaviour, and the project stays simulation-first.

Discrete, frozen choices produce steps

Outage fallback, 1,000 images and frozen operating points give abrupt changes and a few non-monotone points. We did not smooth them.

SPEC §2 treats both a crossover and learned dominance as complete Tier 1 outcomes. Neither result undoes the other.

Four times we changed the approach because of what we found

The rubric asks what we decided based on the results obtained. These are the four that count.

AM-34 · DEC-16

The baseline was capped at QPSK, which made a crossover arithmetically impossible. We added 16-QAM and made modulation adaptive per SNR, so the baseline transmits more as the link improves.

The rule we adopted: every lever strengthens the baseline or is preregistered. We never weaken the learned system.

AM-52

Our SNR grid was sampled every 2 dB exactly where the BPSK waterfall sits, so the classical cliff could have been smeared or missed. We added -7, -5 and -3 dB before the curve mattered.

Dense where the effect lives, not where sampling was convenient.

AM-58

The packetisation checker reported zero failures while violating four of its own rules, including byte alignment and exact code-block division. We fixed the checker, not the results.

All 215 configurations stayed feasible and every scientific outcome was unchanged.

AM-60

Test access was set to release at G-10, which sits in week 9. That opened the test split about three weeks before the freeze manifest existed.

The release moved to G-12 in week 11. It has stayed sealed since.

Twenty-two further amendments are recorded in SPEC §17, including three checks that were wrong and had to be repaired.

What we claim as new, and what we do not

Two claims, both tied to the experimental design rather than to the idea of semantic communication.

ER-9 attribution decomposition

A task-aware digital arm sends learned features over the same LDPC and modulation chain at the same channel-use budget. It separates task-aware representation from joint coding inside one experiment.

Distinguished from: work that reports a learned-versus-classical gap without such a control.

BR-11 format-overhead-controlled baseline

Codec container, framing, CRC, segmentation and rate-matching overhead are measured, charged and published, so the classical arm is tuned on the payload it can actually use.

Distinguished from: pipelines that quote a nominal ratio and leave the overhead unaccounted.

NOT CLAIMED AS NEW

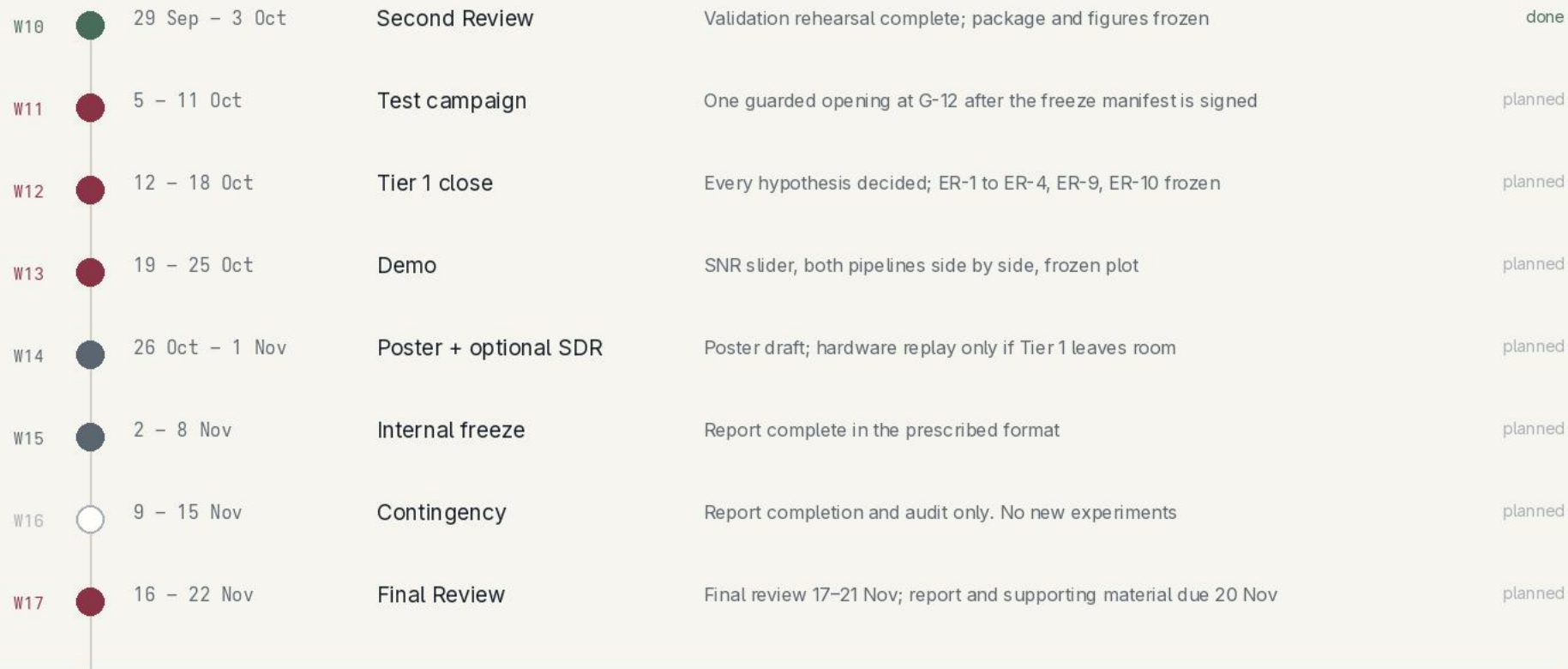
- Graceful degradation under noise
- Avoiding the digital cliff
- A task-aware digital semantic system
- DeepJSCC as a method

All four have substantial prior art. Claiming any of them would cost more than it earns.

NEGATIVE CHECK

Each claim above is checked against the 30 references in docs/literature-review.md, and each negative assertion in the novelty statement must resolve to that list rather than to this specification. The full PR-7 statement ships with the third review package.

Where the remaining weeks went



The report deadline falls inside Final Review week, which is why W15 carries an internal freeze and W16 holds no experimental scope.

Where this leaves the project

A learned system trained for the task keeps answering when the link is too poor for the digital chain to deliver anything. Once the digital chain delivers, it is the more accurate system, and the curves cross between -5 and -4 dB.

Next

One guarded test campaign at G-12, once the freeze manifest is signed. Test access is 0 today.

Then

Every hypothesis decided on the test split, and Tier 1 frozen at G-5.

Later

Tier 2 SDR replay and the live demo are stretch. The project does not depend on them.

EVIDENCE TRAIL

252-unit aggregate and per-image manifest in `results/learned/w10/` · figures and CSVs in `presentation-results/` · requirements and amendments in `spec/SPEC.md` §17

The test split has not been opened. `test_access = 0` across every gate, every run and every artifact in this project.